

Reinforcement Learning for Reducing Spider-Web Structures in ZX-Calculus Optimization

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The ZX-calculus provides a powerful graphical framework for quantum circuit optimization, offering reductions in T-count and CNOT-depth that often exceed gate-level compilers. However, aggressive ZX rewriting frequently produces dense, cyclic clusters of spiders informally called *spider webs*. Such regions are difficult to simplify, tend to destroy gFlow/cFlow, and can lead to exponential blowups during circuit extraction. As a result, post-optimization circuits may exhibit worse 2-qubit costs despite successful local simplifications.

Recent work (e.g., Fischbach et al. 2025) surveys reinforcement-learning (RL) approaches for ZX optimization, but no existing method explicitly targets spider-web formation. This project proposes to develop an RL-guided optimization framework designed to detect, reduce, and avoid spider-web structures in ZX diagrams.

Hypothesis

An RL agent trained to minimize spider-webs will reduce spider-web complexity relative to PyZX `full_reduce` and heuristic baselines.

Methodology

The RL agent selects ZX rewrite rules, including:

- flow-preserving rewrites: local complementation, pivoting, neighbor-unfusion, angle-safe rules;
- optionally, “risky” rewrites (e.g., bialgebra) with explicit penalties for flow loss.

Benchmarks

Structural metrics (amount of spider-Webs, H-edges, cluster sizes, presence of gFlow/cFlow) and compiler-level metrics (post-extraction CNOT count, 2-qubit depth, T-count/T-depth, extraction success/time).